

[illegible]

- Add new screen

```

DCS
Collaborative robot

Force Monitor:   Current      Peak
                  0.00[N] (    0%)    0%

Settings :       Warning      Output
                  0%           GO[  0]

Payload Monitor: Current      Peak
Force           :            0%    0%
Moment          :            0%    0%
Settings :       Warning      Output
Force           :            50%   GO[  0]
Moment          :            50%   GO[  0]
Warning in DSDL:                DISABLE
  
```

DCS				
Collaborative robot				
Force Monitor:				
	Resultant	X	Y	Z
Curr:	0.00[N]			
	0%	0%	0%	0%
Peak:	0%	0%	0%	0%
Output:	GO[0] GO[0]	GO[0] GO[0]	GO[0] GO[0]	GO[0] GO[0]
Warning:	0%			
Payload Monitor:				
FORCE	Resultant	X	Y	Z
Curr:	0%	0%	0%	0%
Peak:	0%	0%	0%	0%
Output:	GO[0] GO[0]	GO[0] GO[0]	GO[0] GO[0]	GO[0] GO[0]
Warning:	50%			
MOMENT	Resultant	X	Y	Z
Curr:	0%	0%	0%	0%
Peak:	0%	0%	0%	0%
Output:	GO[0] GO[0]	GO[0] GO[0]	GO[0] GO[0]	GO[0] GO[0]
Warning:	50%			
Warning in DSBL:		DISABLE		
[TYPE]	CONFIRM	PEAKCLR		UNDO

- Add the periodical check sample for Small Collaborative Robot

The following is the example of the proper position data for the periodical function check.

					Title	Add description for FANUC Robot series R-30iB/R-30iB Mate CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function)				
						Draw No.	B-83744EN/01-03			
Ed.	Date	Design	Description			FANUC CORPORATION			Page	2/6
	Date	2017.4.4	Design.	Y.Naito	Apprv.					

The diagram illustrates a sequence of four poses for a robotic arm, connected by blue arrows indicating the progression of movement. The arm is shown in a green color. The sequence starts with the arm in a vertical position, then moves to a horizontal position, then to a position where the arm is bent at the elbow, and finally to a position where the arm is bent at the elbow and the end effector is pointing downwards.

J3 to J6 = 0 deg

J3 to J6 = 0 deg

J3 to J6 = 0 deg

J3 to J6 = 0 deg

CR-4*i*A, CR-7*i*A, CR-7*i*A/L



J3 to J6 = 0 deg

J3 to J6 = 0 deg

J3 to J6 = 0 deg

J3 to J6 = 0 deg

- Correct the specification of each robot model (underlined part)

Correct

- When a human is in the Zone 1, the contact stop is enabled.
- When a human is in the Zone 2 (including Zone 1), the override is set to 33 % by Override Select function.

[illegible]

-
- Area Sensor
- Zone 1 (Override 33%, Contact stop is enabled)
- Zone 2 (Override 33%, Contact stop is disabled)
- Outside (Override 100%, Contact stop is disabled)

- | | | | |
|-----------|--------|---------|----|
| @Limit 1: | 105.00 | SPI[1] | OK |
| Limit 2: | 0.00 | ---[0] | OK |
| Limit 3: | 0.00 | ---[0] | OK |
| Limit 4: | 0.00 | ---[0] | OK |

- ```

VERRIDE SELECT

1 Function Enable :ENABLE

2 Signal 1 : DI[20][OFF]
3 Signal 2 : DI[21][OFF]

Signal 1 Signal 2 Override
4 OFF OFF 33 %
5 OFF ON 33 %
6 ON OFF 100 %
7 ON ON 100 %

```

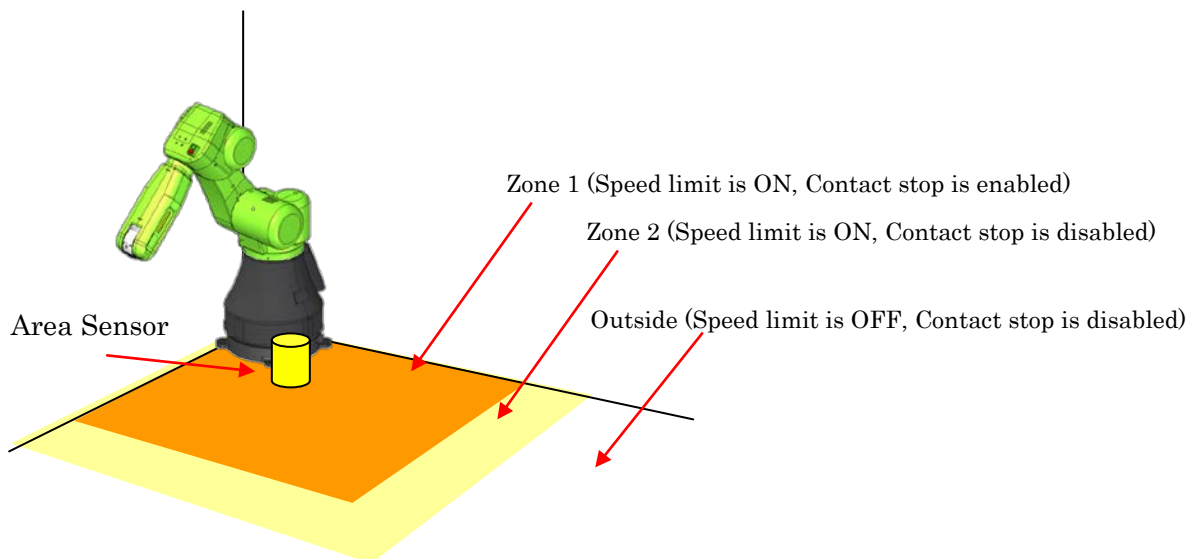
[illegible]



Correct

Correct

- When a human is in the Zone 1, the contact stop is enabled.
- When a human is in the Zone 2 (including Zone 1), the speed limit is ON by turning OFF the disabling input of the speed limit.
- When a human is not in Zone 2 (including Zone 1), the speed limit is OFF by turning ON the disabling input of the speed limit.



- Area sensor has 1 safety output and 1 non-safety output.
- The safety output of the area sensor is connected to SFDI1 of the robot controller. This signal is accessed as SPI[1] in DCS function. When a human is in Zone 1, the SPI[1] becomes OFF.
- The non-safety output of the area sensor is connected to digital input signal that is assigned to DI[20]. When a human is in the Zone 2, DI[20] becomes OFF.
- Set the disabling input of the force limit to SPI[1]. Refer to Section 2.2.

|           |        |         |    |
|-----------|--------|---------|----|
| @Limit 1: | 105.00 | SPI[ 1] | OK |
| Limit 2:  | 0.00   | ---[ 0] | OK |
| Limit 3:  | 0.00   | ---[ 0] | OK |
| Limit 4:  | 0.00   | ---[ 0] | OK |

- Assign the disabling input of the speed limit to DI[20]. Set following system variables.  
`$DCSS_CLLB[1].$DITYP_HISPD = 1` (Type of input DI:1, F:35)  
`$DCSS_CLLB[1].$DIIDX_HISPD = 20` (Index of input)

[illegible]



**⚠ WARNING**  
When the contact stop is disabled, the collaborative robot does not stop even though the external force exceeds the limit, and a serious personal injury could result. If the robot system is designed to be able to disable the contact stop, adequate risk assessment for the whole robot system is necessary to verify that the contact stop can be disabled.

|     |      |          |             |         |        |  |  |                                                                                                                                |                             |                   |      |     |
|-----|------|----------|-------------|---------|--------|--|--|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------|------|-----|
|     |      |          |             |         |        |  |  | Title<br>Add description for FANUC Robot series R-30iB/R-30iB Mate CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function) | Draw No.<br>B-83744EN/01-03 | FANUC CORPORATION | Page | 6/6 |
|     |      |          |             |         |        |  |  |                                                                                                                                |                             |                   |      |     |
|     |      |          |             |         |        |  |  |                                                                                                                                |                             |                   |      |     |
|     |      |          |             |         |        |  |  |                                                                                                                                |                             |                   |      |     |
|     |      |          |             |         |        |  |  |                                                                                                                                |                             |                   |      |     |
|     |      |          |             |         |        |  |  |                                                                                                                                |                             |                   |      |     |
| Ed. | Date | Design   | Description |         |        |  |  |                                                                                                                                |                             |                   |      |     |
|     | Date | 2017.4.4 | Design.     | Y.Naito | Apprv. |  |  |                                                                                                                                |                             |                   |      |     |